Summary in Graph

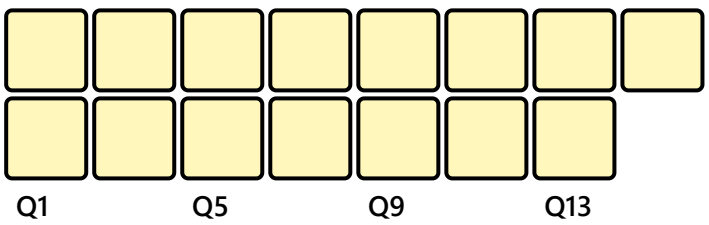
Exam Summary (GO Classes CS | CO and Architecture | Cache Memory | Weekly Quiz 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (20) out of 202 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 CO & Architecture

Match the following types of cache misses with the most appropriate technique to reduce them:

List – I (Cache Misses)

A. Capacity Misses
B. Compulsory Misses

C. Conflict Misses

List - II (Techniques to reduce them)

I. Increase Associativity
II. Increase Block Size
III. Increase Cache Size

A. A-III, B-II, C-I

B. A-II, B-III, C-I

C. A-I, B-II, C-III

D. A-III, B-I, C-II

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | CO & Architecture

Compared to an 8 K-byte direct-mapped cache, what type of misses will a 16 K-byte direct-mapped cache have fewer of? (Mark all that apply).

A. compulsory

B. conflict

C. capacity

D. none

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | CO & Architecture

Let A1 represents an event of playing a video file by accessing the pixel contents of adjacent frames of the video from memory. And let A2 represents an event of initializing a 1D array with the value zero on all array elements sequentially using a for loop. Which of the following is TRUE?

A. Neither A1 nor A2 shows temporal or spatial locality.

B. A1 shows temporal locality and A2 shows spatial locality.

C. Both A1 and A2 show spatial locality.

D. A1 shows spatial locality and A2 shows temporal locality.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | CO & Architecture

A 128 KB n-way set associative cache has a block size of 256 bytes. The cache has 64 sets. Then the value of n is _____

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Consider a machine with a byte-addressable main memory of 2^{20} bytes, a block size of 16 bytes and a direct-mapped cache having 2^{12} cache lines. Let the addresses of two consecutive bytes in main memory be $(E201F)_{16}$ and $(E2020)_{16}$. What are the tag and cache line addresses (in hex) for main memory address $(E201F)_{16}$?

A. E, 201

B. F, 201

C. E, E20

D. 2, 01F

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

In a 32 K-byte four-way set-associative cache with 32byte blocks, how large is the index field used to address the cache array? (write down the number of bits)

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Assume you are given a 64 byte direct-mapped cache. The cache uses a 16 byte line size. Assume that you have a memory reference address stream that is listed below.

Reference Address
0×0000
0×0148
$0 \times 028c$
$0 \times 03c0$
0×0004
0×0408
$0 \times 014c$
$0 \times 000c$

The cache starts in the initial state of all lines invalid. What is the total number of cache misses that occur during this sequence?

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Assume you are given a 64 byte 4-way set associative cache with LRU replacement. The cache uses a 16 byte line size. Assume that you have a memory reference address stream that is listed below.

Reference Address
0×0000
0×0148
$0 \times 028c$
$0 \times 03c0$
0×0004
0×0408
$0 \times 014c$
$0 \times 000c$

The cache starts in the initial state of all lines invalid. What is the total number of cache misses that occur during this sequence?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 01sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider the following two questions:

1. Can a direct-mapped cache sometimes have a higher hit rate than a fully associative cache with an LRU replacement policy (on the same memory reference pattern and with the same cache size)?

2. Can a 2-way set cache sometimes have a higher hit rate than a fully associative cache with an LRU replacement policy (on the same memory reference pattern and with the same cache size)?

What's the correct answer?

A. Yes, Yes

B. Yes, No

C. No, Yes

D. No, No

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | CO & Architecture

The following problem concerns basic cache lookups.

- The memory is byte-addressable.
- Memory accesses are to 1-byte words (not 4-byte words).
- Physical addresses are 13 bits wide.
- The cache is 4-way set associative, with a 4-byte block size and 32 total lines.

In the following tables, all numbers are given in hexadecimal. The Index column contains the set index for each set of 4 lines. The Tag columns contain the tag value for each line. The V column contains the valid bit for each line. The Bytes 0-3 columns contain the data for each line, numbered left-to-right starting with byte 0 on the left.

The contents of the cache are as follows:

4-way Set Associative Cache																								
Index	Tag	V	Bytes 0-3				Tag	V	Bytes 0-3				Tag	V	Bytes 0-3				Tag	V	Bytes 0-3			
0	84	1	ED	32	0A	A2	9E	0	BF	80	1D	FC	10	0	EF	9	86	2A	E8	0	25	44	6F	1A
1	18	1	03	3E	CD	38	E4	0	16	7B	ED	5A	02	0	8E	4C	DF	18	E4	1	FB	B7	12	02
2	84	0	54	9E	1E	FA	84	1	DC	81	B2	14	48	0	B6	1F	7B	44	89	1	10	F5	B8	2E
3	92	0	2F	7E	3D	A8	9F	0	27	95	A4	74	57	1	07	11	FF	D8	93	1	C7	B7	AF	C2
4	84	1	32	21	1C	2C	FA	1	22	C2	DC	34	73	0	BA	DD	37	D8	28	1	E7	A2	39	BA
5	A7	1	A9	76	2B	EE	73	0	BC	91	D5	92	28	1	80	BA	9B	F6	6B	0	48	16	81	0A
6	8B	1	5D	4D	F7	DA	29	1	69	C2	8C	74	B5	1	A8	CE	7F	DA	BF	0	FA	93	EB	48
7	84	1	04	2A	32	6A	96	0	B1	86	56	0E	CC	0	96	30	47	F2	91	1	F8	1D	42	30

For which of the given physical addresses does a cache miss occur?

A. Physical address: $0x0D74$

B. Physical address: $0x0AEE$

C. Physical address: $0x0A2E$

D. Physical address: $0x109C$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

The following problem concerns basic cache lookups.

- The memory is byte-addressable.
- Memory accesses are to 1-byte words (not 4-byte words).
- Physical addresses are 13 bits wide.
- The cache is 4-way set associative, with a 4-byte block size and 32 total lines.

In the following tables, all numbers are given in hexadecimal. The Index column contains the set index for each set of 4 lines. The Tag columns contain the tag value for each line. The V column contains the valid bit for each line. The Bytes 0-3 columns contain the data for each line, numbered left-to-right starting with byte 0 on the left.

The contents of the cache are as follows:

4-way Set Associative Cache																								
Index	Tag	V	Bytes 0-3				Tag	V	Bytes 0-3				Tag	V	Bytes 0-3				Tag	V	Bytes 0-3			
0	84	1	ED	32	0A	A2	9E	0	BF	80	1D	FC	10	0	EF	9	86	2A	E8	0	25	44	6F	1A
1	18	1	03	3E	CD	38	E4	0	16	7B	ED	5A	02	0	8E	4C	DF	18	E4	1	FB	B7	12	02
2	84	0	54	9E	1E	FA	84	1	DC	81	B2	14	48	0	B6	1F	7B	44	89	1	10	F5	B8	2E
3	92	0	2F	7E	3D	A8	9F	0	27	95	A4	74	57	1	07	11	FF	D8	93	1	C7	B7	AF	C2
4	84	1	32	21	1C	2C	FA	1	22	C2	DC	34	73	0	BA	DD	37	D8	28	1	E7	A2	39	BA
5	A7	1	A9	76	2B	EE	73	0	BC	91	D5	92	28	1	80	BA	9B	F6	6B	0	48	16	81	0A
6	8B	1	5D	4D	F7	DA	29	1	69	C2	8C	74	B5	1	A8	CE	7F	DA	BF	0	FA	93	EB	48
7	84	1	04	2A	32	6A	96	0	B1	86	56	0E	CC	0	96	30	47	F2	91	1	F8	1D	42	30

For the given physical address below, indicate the cache byte value returned in hex.

Physical address: 0x0AEE

A. FA

B. FF

C. 11

D. It's a miss

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

The following problem concerns basic cache lookups.

- The memory is byte addressable.
- Memory accesses are to 1-byte words (not 4-byte words).
- Physical addresses are 13 bits wide.
- The cache is 4-way set associative, with a 4-byte block size and 32 total lines.

In the following tables, all numbers are given in hexadecimal. The Index column contains the set index for each set of 4 lines. The Tag columns contain the tag value for each line. The V column contains the valid bit for each line. The Bytes 0-3 columns contain the data for each line, numbered left-to-right starting with byte 0 on the left.

The contents of the cache are as follows:

4-way Set Associative Cache																								
Index	Tag	V	Bytes 0-3				Tag	V	Bytes 0-3				Tag	V	Bytes 0-3				Tag	V	Bytes 0-3			
0	84	1	ED	32	0A	A2	9E	0	BF	80	1D	FC	10	0	EF	9	86	2A	E8	0	25	44	6F	1A
1	18	1	03	3E	CD	38	E4	0	16	7B	ED	5A	02	0	8E	4C	DF	18	E4	1	FB	B7	12	02
2	84	0	54	9E	1E	FA	84	1	DC	81	B2	14	48	0	B6	1F	7B	44	89	1	10	F5	B8	2E
3	92	0	2F	7E	3D	A8	9F	0	27	95	A4	74	57	1	07	11	FF	D8	93	1	C7	B7	AF	C2
4	84	1	32	21	1C	2C	FA	1	22	C2	DC	34	73	0	BA	DD	37	D8	28	1	E7	A2	39	BA
5	A7	1	A9	76	2B	EE	73	0	BC	91	D5	92	28	1	80	BA	9B	F6	6B	0	48	16	81	0A
6	8B	1	5D	4D	F7	DA	29	1	69	C2	8C	74	B5	1	A8	CE	7F	DA	BF	0	FA	93	EB	48
7	84	1	04	2A	32	6A	96	0	B1	86	56	0E	CC	0	96	30	47	F2	91	1	F8	1D	42	30

For the given contents of the cache, how many physical memory addresses will hit in Set 7?

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

The following problem concerns basic cache lookups.

- The memory is byte-addressable.
- Memory accesses are to 1-byte words (not 4-byte words).
- Physical addresses are 13 bits wide.
- The cache is 4-way set associative, with a 4-byte block size and 32 total lines.

In the following tables, all numbers are given in hexadecimal. The Index column contains the set index for each set of 4 lines. The Tag columns contain the tag value for each line. The V column contains the valid bit for each line. The Bytes 0-3 columns contain the data for each line, numbered left-to-right starting with byte 0 on the left.

The contents of the cache are as follows:

4-way Set Associative Cache																								
Index	Tag	V	Bytes 0-3				Tag	V	Bytes 0-3				Tag	V	Bytes 0-3				Tag	V	Bytes 0-3			
0	84	1	ED	32	0A	A2	9E	0	BF	80	1D	FC	10	0	EF	9	86	2A	E8	0	25	44	6F	1A
1	18	1	03	3E	CD	38	E4	0	16	7B	ED	5A	02	0	8E	4C	DF	18	E4	1	FB	B7	12	02
2	84	0	54	9E	1E	FA	84	1	DC	81	B2	14	48	0	B6	1F	7B	44	89	1	10	F5	B8	2E
3	92	0	2F	7E	3D	A8	9F	0	27	95	A4	74	57	1	07	11	FF	D8	93	1	C7	B7	AF	C2
4	84	1	32	21	1C	2C	FA	1	22	C2	DC	34	73	0	BA	DD	37	D8	28	1	E7	A2	39	BA
5	A7	1	A9	76	2B	EE	73	0	BC	91	D5	92	28	1	80	BA	9B	F6	6B	0	48	16	81	0A
6	8B	1	5D	4D	F7	DA	29	1	69	C2	8C	74	B5	1	A8	CE	7F	DA	BF	0	FA	93	EB	48
7	84	1	04	2A	32	6A	96	0	B1	86	56	0E	CC	0	96	30	47	F2	91	1	F8	1D	42	30

For the given contents of the cache, what is the probability (expressed as a percentage) of a cache hit when the physical memory address ranges between 0x1080 and 0x109F? Assume that all addresses are equally likely to be referenced.

Your Answer: | Correct Answer: 50 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

The following 13 block requests $\mathrm{A}, \mathrm{B}, \mathrm{A}, \mathrm{C}, \mathrm{D}, \mathrm{C}, \mathrm{B}, \mathrm{A}, \mathrm{C}, \mathrm{D}, \mathrm{E}, \mathrm{F}, \&\mathrm{E}$ are mapped to set n of a 2-way set associative cache memory that uses LIFO block replacement policy. Assume that set n is initially empty; then the total number of misses encountered after servicing all these requests is _____.

Your Answer: | Correct Answer: 11 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

The address of a word in a byte-addressable 1 MB physical memory is $0\mathrm{xB}7496$. This word, upon being brought to the cache, is mapped to set number 36 (set numbering starts from 0 to $n-1$). The block size of the cache memory (in bytes) is _____.

Your Answer: | Correct Answer: 32 | Not Attempted | Time taken: 00min 00sec | Discuss

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